NetStrike

Simple Routed Network Simulation with Client, Attacker, and 3Web Server in GNS

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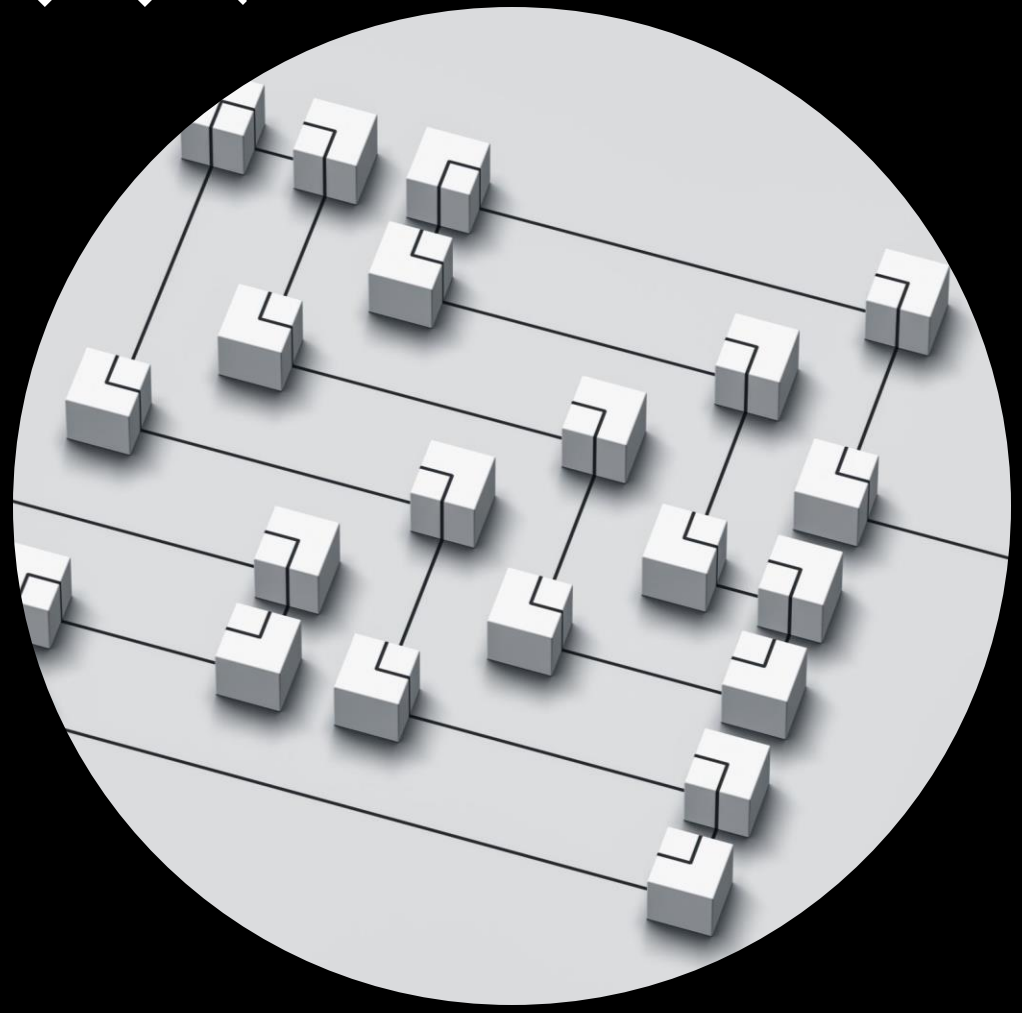


Introduction

- NetStrike is a small GNS3 network simulation project.
- It includes a Kali Linux attacker, a legitimate web client, one router, one switch, and a web server.
- The goal is to demonstrate routed communication between local hosts and a remote server.
- The project also shows basic attacker-side interaction using Kali Linux.

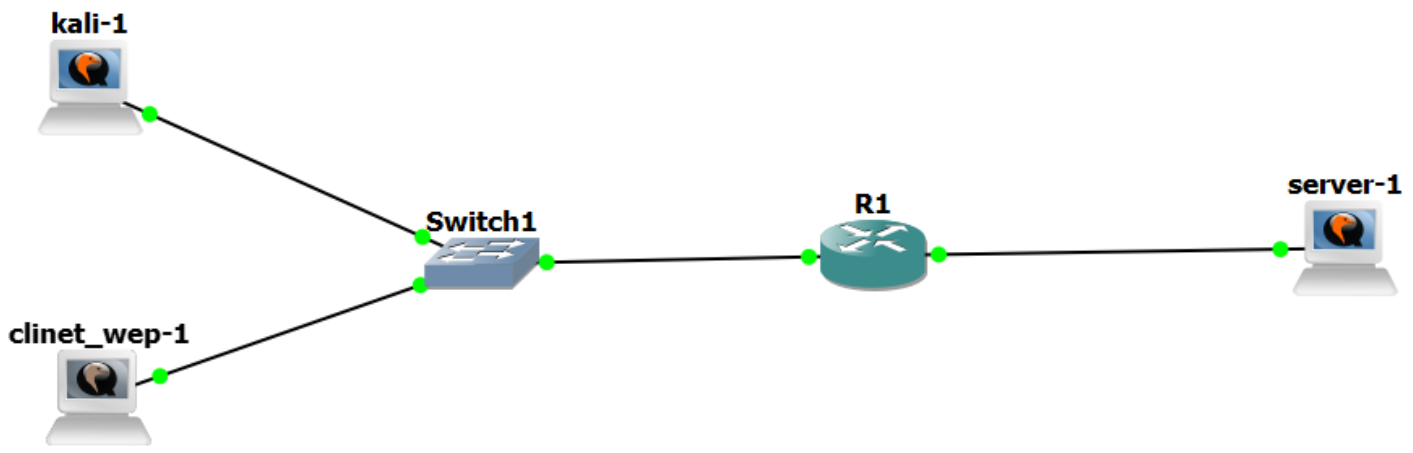
Problem Statement

The client and attacker are placed on one local subnet.

The server is placed on a separate subnet behind router R1.

Communication must pass through the router instead of staying in one flat LAN.

Successful access depends on correct IP addressing, default gateways, and routing.



Topology

Kali / Client → Switch1 → Router R1 → Server

```
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP grou
p default qlen 1000
    link/ether 0c:c5:2a:d2:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet6 fe80::ec5:2aff:fed2:0/64 scope link
       valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$ sudo ip add addr 192.168.1.20/24 dev ens3
Command "addr" is unknown, try "ip address help".
ubuntu@ubuntu:~$ sudo ip addr add 192.168.1.20/24 dev ens3
ubuntu@ubuntu:~$ sudo ip route add default via 192.168.1.1
ubuntu@ubuntu:~$ ping 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=255 time=204 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=255 time=18.7 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=255 time=12.4 ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=255 time=15.8 ms
```

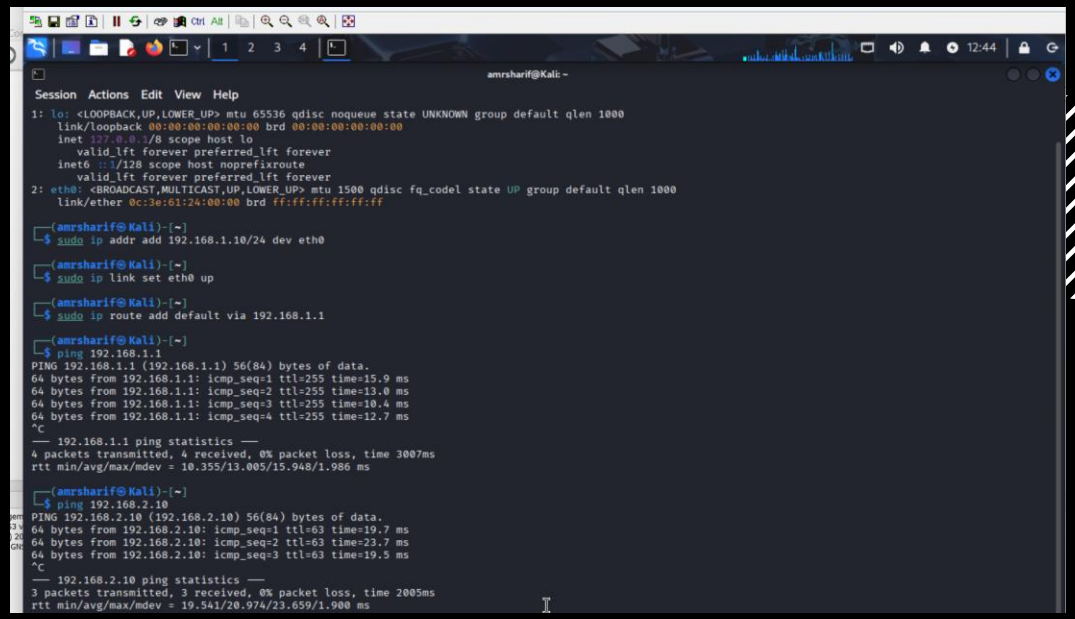
```
ubuntu@ubuntu: ~/site
ubuntu@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP grou
p default qlen 1000
    link/ether 0c:fb:fa:f3:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
ubuntu@ubuntu:~$ sudo addr add 192.168.2.10/24 dev ens3
sudo: addr: command not found
ubuntu@ubuntu:~$ sudo ip addr add 192.168.2.10/24 dev ens3
ubuntu@ubuntu:~$ sudo ip route add default via 192.168.2.1
ubuntu@ubuntu:~$ mkdir ~/site
ubuntu@ubuntu:~$ echo "<h1>Netstrike Project</h1>" > ~/site/index.html
ubuntu@ubuntu:~$ cd ~/site
ubuntu@ubuntu:~/site$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
```

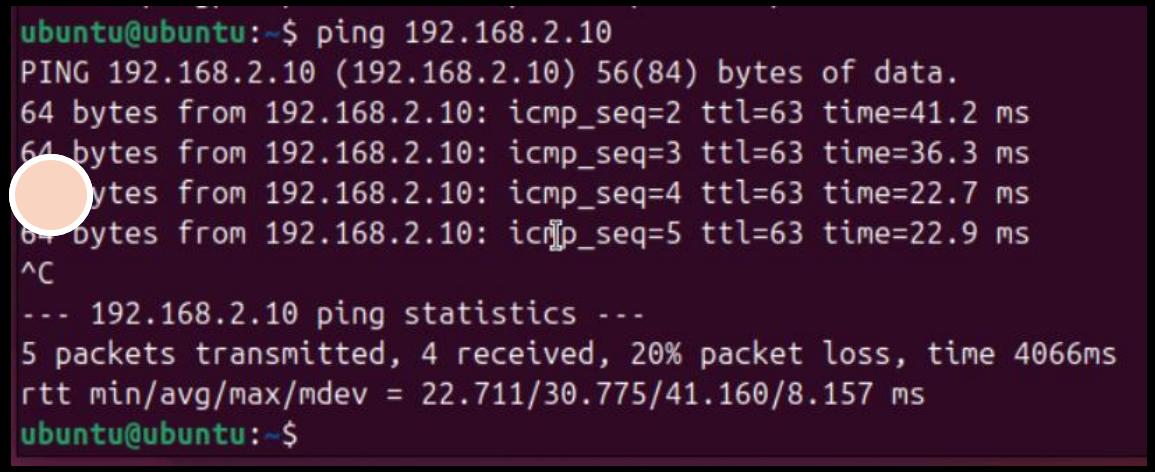
IP Addressing Scheme

- kali-1 → 192.168.1.10
- clinet_wep-1 → 192.168.1.20
- R1 f0/0 → 192.168.1.1
- R1 f0/1 → 192.168.2.1
- server-1 → 192.168.2.10

Testing and Verification

- The client successfully pinged the web server.
- Kali Linux successfully reached both the router and the server.
- These tests confirmed end-to-end connectivity through R1





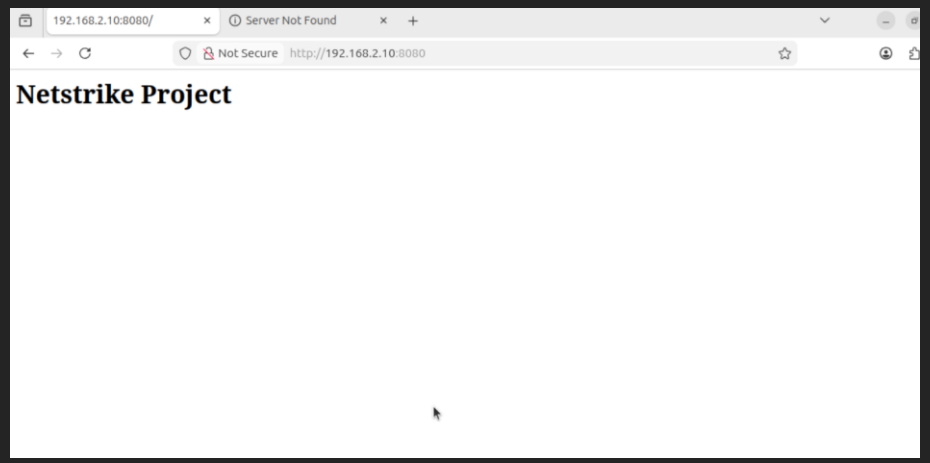
```
┌──(amrsharif㉿Kali)-[~]
└─$ nmap -Pn -n -p 8080 192.168.2.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-10 12:45 +0300
Nmap scan report for 192.168.2.10
Host is up (0.019s latency).

PORT     STATE SERVICE
8080/tcp open  http-proxy

Nmap done: 1 IP address (1 host up) scanned in 0.28 seconds
```

Security Perspective

- Kali Linux represents the attacker node in the topology.
- It was used for reconnaissance and service verification.
- Nmap confirmed that port 8080 was open on the server.

Results

- The web server successfully hosted the project page.
- The client accessed the page through the browser.
- The topology proved correct routing, gateway assignment, and service delivery.



Conclusion

- NetStrike demonstrates a simple routed client-server environment in GNS3.
- The project combines networking fundamentals with basic attacker-side testing.
- It provides a clear model for routing, host configuration, and introductory security validation.

Questions? •

NetStrike Project •

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Thank You